Fall 2025

CSE 360: Introduction to Software Engineering.

1. Contact Information

INSTRUCTOR: Dr. Zahra Sadri-Moshkenani, Ph.D.
Assistant Teaching Professor

E-MAIL: zsadrimo@asu.edu

OFFICE: Brickyard 418

OFFICE HOURS: Check out Module 0 -> Office Hours or by appointment

2. Instructional Team Office Hours

Office hours for the instructions team and how to find and contact them can be found on Canvas in Module 0: Welcome! Start Here! (Teaming Team and Office Hours).

3. Course Description:

Introduces principles of software engineering, including software life cycle models, hands-on experience using selected software engineering tools, agile programming, software project management, continuous integration and development, software testing and quality, software architecture, information management systems, teamwork, legal and ethical issues, and emerging technologies such as artificial intelligence.

4. Enrollment Requirements

CSE 220 or CSE 240 with C or better OR Computer Science or Software Engineering graduate student OR Visiting University Student; Biomedical Informatics BS or Computer Science BS or Digital Culture (Media Processing) BA/BS or Geographic Information Science BS or Computer Systems BSE or Engineering Management BSE major.

If you don't meet the official prerequisites but are admitted into the course because you did courses *"equivalent"* to the official prerequisite courses, it is **your** responsibility to make sure you understand the necessary background material and can use your understanding in the course activities, assignments, quizzes, and examinations.

5. Expected Learning Outcomes

- Explain software engineering fundamentals (all modules)

- Develop use-cases and use-case diagrams to elicit software requirements using a current modeling tool (Modules 4, 5, 6, 9)
- Analyze and design software systems using object-oriented techniques (Modules 7, 8, 9)
- Develop code by applying proper object-oriented coding techniques and applying suitable design patterns (Modules 5, 6, and 7)
- Perform functional and unit testing of software systems using state-of-the-art testing tools and techniques (Modules 10 and 11)
- Follow a disciplined software process model in developing a software product (Modules 2 and 3)
- Work effectively on a software development team (Module 2)

6. ABET Computer Science Student Outcome and Assessment

CS (2) Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline -- **Assessed**

CS (3) Communicate effectively in a variety of professional contexts.

CS (5) Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline. - **Assessed**

CS (6) Apply computer science theory and software development fundamentals to produce computing-based solutions.

7. Barrett Honors Enrichment Contracts

This course supports students wishing to earn honors credit from the Barrett Honors College. To learn more about this program, please read the Canvas Module with the same title.

Students wishing to earn honors credit must propose a project that extends some practical aspect of the Team Project being done by the student's team. Even though it extends the team's project, the extension must be done individually without input from or impact on the rest of the student's team.

If you are interested in this opportunity, please schedule an appointment with your professor within the first two weeks of the start of the course. A Barrett Honors Contract for CSE 360 **requires regular short check-ins and status updates with the professor**.

8. Grade Policies

The following is the grading scale that will be used in the course:

- A+ >= 97% and <= 100%
- A >= 90% and < 97%
- B+ >= 87% and < 90%
- B >= 80% and < 87%
- C+ >= 77% and < 80%
- C >= 70% and < 77%
- D >= 60% and < 70%
- E < 60%

Grades will **not be curved** and will **not be rounded up**. Attempting to influence or change any Academic Evaluation or academic record for reasons having no relevance to academic achievement (e.g., to help you avoid losing a scholarship) is **unethical** and will be reported. Grades reflect your performance on assignments and adherence to deadlines. Grades are independent of how well any other student performs in the class.

Grades on assignments will be available within two weeks of the due date in the Gradebook.

Any **discrepancy or disagreement in grading must be presented within one week** of receiving your graded materials; **otherwise, no grade change will be made**. Appealing a grade more than one week after receiving the grade will be ignored, and **no grade change will be made**. Your course grade will be based on the distribution below:

▪ Exams	30%	Midterm (15%), and Final (15%)
▪ Module Quizzes	15%	12 Quizzes
▪ Team Project	30%	3 Assignments
▪ Individual Homework	20%	3 Assignments
▪ Class Participation	5%	Activities during class (2.5%) and during the module (2.5%)

Exams: There will be **two exams** (midterms and a final exam) given during the semester. **All exams are closed book and no other information sources, with one exception, are allowed.** Two pieces of paper with handwritten or printed notes on both sides are permitted, as are blank paper and a pen and pencil.

Module Quizzes: Module quizzes will be based on the prerecorded hybrid lectures for the module. There are **no make-ups** for missed module quizzes.

Team Project: In a team of typically 5 students, students will complete multiple project deliverables, including documents, screencasts, and programs. (Teams may be smaller due to students leaving the course, and may be larger by one in special cases.) These deliverables will be announced in the Team Project section of the course Canvas. All team members are expected to actively and equally participate in the creation of **all deliverables and the submission of those deliverables**. Submitting the "wrong deliverable" is not an excuse for a late submission. The grade for the team project may be decreased if there is inadequate participation from any team member. Deliverables and due dates will be announced and discussed in the lecture and presented in the Team Project section of the course Canvas. **No late submissions will be accepted for the class project (study the "Absence policies…" section below carefully)**. In addition, a report due at the end of the team project will include an evaluation of your team's work, your contributions, as well as those of your teammates. A specific template for the content will be provided.

Individual Homework Assignments: They will be given throughout the semester. Homework assignments involve coding and the use of tools needed for the following Team Project Phase submission. Homework assignments **can never be made up** and **can never be turned in late unless you have a documented and approved ASU-recognized extenuating circumstance (e.g., medical reason)**. You must gain approval from the instructional team **before or within one week** after the submission deadline to arrange for make-up activities due to approved reasons.

9. Submission of Deliverables.

Submission: An assignment is **not submitted** until Canvas shows that the **submission process has been completed before the deadline.** Starting the upload before the deadline is **not sufficient**! The process can take several minutes with large submission files and even longer when many students are trying to submit at the same time (e.g., just before the deadline). If you start the process, but it does not finish before the deadline, you will typically receive a failure message from Canvas.

Multiple Submissions: You are allowed and even encouraged to submit an incomplete assignment early, in case you or the team have problems finishing the submission before the deadline. The last assignment submitted will be the one graded. If a final submission has not been made more than an hour before the deadline, submit what you have. The last few points to be earned by "finishing" the assignment are not worth the zero earned if you fail to submit before the deadline.

Team Project Submissions: The **whole team is responsible** for the Team Project submission being **correct** and **submitted before the deadline**. This is not something that can be delegated to just one team member. A virtual or physical team meeting should be scheduled to perform the submission, so each member of the team can see that the right material is being submitted, and it is completed before the deadline. **Failing to participate** in the final submission means **you approve** of all aspects of the submission, and you will **not dispute the grade earned** should the wrong files be submitted or if the submission fails to complete the process before the deadline.

10. Absence policies and the conditions under which assigned work and/or tests can be made up.

Excused absences for classes will be given without penalty to the grade in the case of (1) a universitysanctioned event [ACD 304-02]; (2) religious holidays [ACD 304-04]; a list of religious holidays can be found here https://eoss.asu.edu/cora/holidays]; (3) work performed in the line-of-duty according [SSM 201-18]. Students who request an excused absence must follow the policy/procedure guidelines. Excused absences do not relieve students of responsibility for any part of the coursework required during the period of absence.

In all cases, **documented evidence must be provided. It must be signed by a doctor, clinic administrator, ASU event organizer, ASU-recognized individual, or a similar official** associated with the event or circumstance, **with the dates to be excused**. In this syllabus, when the phrase **documented ASU-recognized extenuating circumstance** is used, we refer to this. If you are sick, a photo of a positive COVID or other home test **is not adequate documentation**. Please visit an ASU clinic, your doctor, or some other healthcare facility and obtain a written statement about any medical condition **and the period** that you should be excused from schoolwork. If you are too sick to go to a clinic, dial 911 and get help!

Homework assignments **can never be made up** and **can never be turned in late without a properly documented ASU-recognized extenuating circumstance**.

Excused absences only apply to **individual assignments**. Excused absences **do not apply to team project assignments**. If you are unable to contribute to your team project due to an instructionalteam-approved excused absence, you need to negotiate with your team to find a mutually acceptable accommodation. They will need to complete the project without your help if you can't honor your commitments on time.

11. Faculty recording of class sessions

Faculty may record class meetings to make an archived recording available to enrolled students with properly documented extenuating circumstances or SAILS accommodations. These recordings will capture just the slides and the voice of the lecturer.

12. Textbook and Recommended Reference Materials

Required Textbook: Software Engineering, 10th Edition, Ian Sommerville, Pearson, 2016 ISBN-13: 978-0-13-394303-0

This textbook is automatically made available to you in a digital form inside the course Canvas. You will be charged for this digital book. You may choose to opt out (see details below).

The required material for this course, CSE 360: Introduction to Software Engineering, will be provisioned as an e-book and made available at a discounted price, cheaper than if purchased directly from the publisher. If you wish to take advantage of this discounted price, no additional action is needed. Following the drop/add deadline, a charge of $50.75, plus tax, will post to your student account under the header "Digital Integrated Course Mtrl" and your access will continue uninterrupted.

If you'd rather purchase the material from an alternate source, you may choose to opt out of the program by using a campus-specific link similar to this: https://includedcp.follett.com/1230†. Enter your ASU e-mail address AS IT APPEARS IN THE ASU DIRECTORY (http://asu.edu/directory), then follow the instructions provided. Be aware that if you do opt out, your access to the e-book will be discontinued.

TO ACCESS THE E-BOOK, CLICK ON THE ASU BOOKSTORE TOOL IN YOUR COURSE'S CANVAS SHELL (you might need to disable pop-up blockers)

Please note: the ebook won't appear on your shelf until approximately 5 days prior to the start of classes. If you need assistance accessing the book or the opt-out portal, fill out the support request form: https://forms.gle/uD4GhBxMoixnbwYx5 †The link to use is campus specific:

- Tempe/Online: https://includedcp.follett.com/1230
- Poly: https://includedcp.follett.com/1232
- West: https://includedcp.follett.com/1233
- Downtown: https://includedcp.follett.com/1234

Optional Textbook: Unified Modeling Language Standards, http://www.uml.org/ (Free)

13. Computer Requirements

This course requires a computer with internet access and the following technologies:

- A Mac or Windows **laptop** is highly recommended to perform in-class activities. If you do not have such a laptop, please schedule an appointment with your professor to discuss this situation as soon as possible.
- Web browsers (Chrome, Mozilla Firefox, or Safari)
- Adobe Acrobat Reader (free)

- Webcam, microphone, headset/earbuds, and speaker
- Microsoft Office (Microsoft 365 is free for all currently enrolled ASU students) ● Reliable broadband internet connection (DSL or cable) to stream videos.

14. Required Course Tools

Programming Language for Projects/Individual Assignments:

- Java (**Required**): Please use the most current version that runs on your machine/OS.
- JavaFX (**Required**): Please use the most current version that runs on your machine/OS.
- Eclipse (**Required**): Please use the most current version that runs on your machine/OS.

Project Management (Not required, but highly recommended):

- Jira: (free version)
- Microsoft Project *UML Modeling:*
- Astah (**Required**): UML diagrams drawn using any other tool or hand-drawn UML diagrams will not be accepted. Astah has a free version for students.

Unit Testing:

- JUnit (**Required**) *Project Repository:*
- GitHub (Highly recommended)

15. Course Itinerary

This Hybrid course requires students to attend an in-person class session and study recorded (Hybrid) lectures from each of the weekly course modules. The Hybrid lectures and the textbook readings provide the principles and concepts at the heart of this course. The in-person sessions provide the practical application, best practice, and opportunities to practice both in a supportive environment.

Week 01: Module 01: What is software engineering?
Week 02: Module 02: Software project management and teamwork
Week 03: Module 3: Software Risk Management and Technical Communication
Week 04: Module 04: Software process models
Week 05: Module 05: Understanding software requirements
Week 06: Module 06: Requirement Analysis Part I
Week 07: Module 07: Requirement Analysis Part II
Week 08: Midterm Review and Fall Break (11 - 14 Oct)
Week 09: Midterm Exam
Week 10: Module 08: Introduction to Software Design and Commonly Used Design Patterns
Week 11: Module 09: Software Architecture
Week 12: Module 10: Introduction to Information Management Systems
Week 13: Module 11: Software Testing and Software Quality
Week 14: Module 12: Structural Testing and Technical Reviews
Week 15: Thanksgiving and Final Exam Review

Week 16: Final Exam Week

16. Policy regarding expected student behavior

Students in this class are expected to acknowledge and embrace the FSE student professionalism expectation located at: https://engineering.asu.edu/professionalism/

17. Generative AI

Generative AI is a technology that can often be useful in helping students learn the theories and concepts in this course. However, unless explicitly allowed by your instructor, the use of generative AI tools to complete any portion of a course assignment or exam will be considered academic dishonesty and a violation of the ASU Academic Integrity Policy. Students confirmed to be engaging in nonallowable use of generative AI will be sanctioned according to the academic integrity policy and FSE sanctioning guidelines.

18. Academic Integrity

All students are expected to adhere to the ASU Student Honor Code and the ASU academic integrity policy, which can be found at https://provost.asu.edu/academic-integrity/policy. Students are responsible for reviewing this policy and understanding each of the areas in which academic dishonesty can occur. If you have taken this course before, you may not reuse or submit any part of your previous assignments without the express written permission of the instructor.

All student academic integrity violations are reported to the Fulton Schools of Engineering Academic Integrity Office (AIO). Withdrawing from this course will not absolve you of responsibility for an academic integrity violation and any sanctions that are applied. The AIO maintains a record of all violations and has access to academic integrity violations committed in all other ASU colleges/schools.

Specific academic integrity announcements for this class: It is an Academic Integrity Violation (AIV) to provide components of or entire solutions to course individual assignments to other students, and team assignments to any student outside of the team. Files stored in any publicly-accessible repository (e.g., Google Drive, GitHub) **must have permissions set** to ensure improper access is not possible. Should members of the instructional team see that solutions or their components are accessible to others, it **will be documented and reported to the AIO**, whether or not anyone has accessed them.

19. Student Copyright Responsibilities

You must refrain from uploading to this course shell, discussion board, website used by the course instructor, or any other course forum, material that is not your own original work, unless you first comply with all applicable copyright laws. Course instructors reserve the right to delete materials from the course shell on the grounds of suspected copyright infringement.

The contents of this course, including lectures and other instructional materials, are copyrighted materials. Students may not share outside the class, including uploading, selling, or distributing course content or notes taken during the conduct of the course. Any recording of class sessions by students is prohibited, except as part of an accommodation approved by the Disability Resource Center. (see ACD 304–06, “Commercial Note Taking Services” and ABOR Policy 5-308 F.14 for more information).

20. Policy against threatening behavior, per the Student Services Manual, SSM 104–02

Students, faculty, staff, and other individuals do not have an unqualified right of access to university grounds, property, or services (see SSM 104-02). Interfering with the peaceful conduct of universityrelated business or activities or remaining on campus grounds after a request to **leave may be considered a crime. All incidents and allegations of violent or threatening conduct** by an ASU student (whether on- or off-campus) must be reported to the ASU Police Department (ASU PD) and the Office of the Dean of Students.

21. Disability Accommodations

Suitable accommodations are made for students having disabilities. Students needing accommodation must register with the ASU Student Accessibility and Inclusive Learning Services office and provide documentation of that registration to the instructor. Students should communicate the need for an accommodation in enough time for it to be properly arranged. See ACD 304-08 Classroom and Testing Accommodations for Students with Disabilities.

22. Harassment and Sexual Discrimination

Arizona State University is committed to providing an environment free of discrimination, harassment, or retaliation for the entire university community, including all students, faculty members, staff employees, and guests. ASU expressly prohibits discrimination, harassment, and retaliation by employees, students, contractors, or agents of the university based on any protected status: race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, and genetic information.

Title IX is a federal law that provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy make clear that sexual violence and harassment based on sex are prohibited. An individual who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic support, from the university. If you or someone you know has been harassed on the basis of sex or sexually assaulted, you can find information and resources at https://sexualviolenceprevention.asu.edu/faqs.

As a mandated reporter, I am obligated to report any information I become aware of regarding alleged acts of sexual discrimination, including sexual violence and dating violence. ASU Counseling Services, https://eoss.asu.edu/counseling is available if you wish to discuss any concerns confidentially and privately. ASU online students may access 360 Life Services using the following link: https://goto.asuonline.asu.edu/success/online-resources.html.

23. Photo requirement

Arizona State University requires each enrolled student and university employee to have on file with ASU a current photo that meets ASU's requirements (your "Photo"). ASU uses your Photo to identify you, as necessary, to provide you with educational and related services as an enrolled student at ASU. If you do not have an acceptable Photo on file with ASU, or if you do not consent to the use of your photo, access to ASU resources, including access to course material or grades (online or in person), may be negatively affected, withheld, or denied.

24. Wait for an Absent Instructor

How Long Students Should Wait for an Absent Instructor: In the event the instructor fails to indicate a time obligation, the time obligation will be 15 minutes for CSE 360, as class sessions last 90 minutes or less. Students may be directed to wait longer by someone from the academic unit if they know the instructor will arrive shortly.

25. Syllabus changes

Any information in this syllabus (other than grading and absence policies) may be subject to change with reasonable advance notice.